

# Alexandra Catherine Montgomery-Whitfield

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San Francisco Bay Area, California, United States

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Principal software engineer and engineering leader with 12+ years of experience architecting distributed systems, leading large teams, and shipping consumer and enterprise products at scale. Deep expertise across frontend, backend, infrastructure, and platform engineering. A pragmatic technologist who pairs long-term architectural vision with relentless execution. Reachable at <https://averylongportfoliourlthatdoesnotbreaknicely.example.com/portfolio/work/2024> for a full case-study archive.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Go, Rust, Java, Kotlin, Ruby, C++, SQL

**Frontend:** React, Next.js, Vue, Svelte, Tailwind CSS, Redux, React Query, Webpack, Vite, Storybook

**Backend:** Node.js, Express, NestJS, GraphQL, gRPC, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ

**Cloud and DevOps:** AWS, GCP, Azure, Docker, Kubernetes, Terraform, Pulumi, GitHub Actions, Prometheus, Grafana

**Practices:** Distributed Systems Architecture and Design, Event-Driven Microservices, Domain-Driven Design, Observability, Site Reliability Engineering

**Leadership:** Team Building, Mentorship, Technical Strategy, Cross-functional Collaboration, Hiring and Interviewing

## PROFESSIONAL EXPERIENCE

### Principal Software Engineer and Technical Lead

2021-06 - Present

#### Hyperscale Cloud Infrastructure Corporation

San Francisco, CA

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 2: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing infrastructure cost by a measurable margin and improving p95 latency across critical paths.
- Achievement 5: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 6: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 7: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.

### Staff Software Engineer

2019-01 - 2021-05

#### Meridian Financial Technologies

New York, NY

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 2: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
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- Achievement 5: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 6: shipped, measured, and iterated on a meaningful improvement that moved a key business

metric and was adopted across the wider organisation.

## **Senior Full-Stack Engineer**

2016-08 - 2018-12

### **Nimbus Analytics**

Remote

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
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- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
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- Achievement 6: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.

## **Software Engineer II**

2014-03 - 2016-07

### **BrightPath Software Solutions and Consulting Group International**

Austin, TX

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 2: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing infrastructure cost by a measurable margin and improving p95 latency across critical paths.
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## **Founding Engineer**

2012-09 - 2014-02

### **StartupForge**

Palo Alto, CA

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
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- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing infrastructure cost by a measurable margin and improving p95 latency across critical paths.
- Achievement 5: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.

## **Junior Developer (no achievements listed)**

2011-06 - 2012-08

### **Legacy Systems Inc.**

Boston, MA

## **Engineering Intern**

2010-06 - 2010-09

### **Internship Co.**

Seattle, WA

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
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- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing

infrastructure cost by a measurable margin and improving p95 latency across critical paths.

## SELECTED PROJECTS

### Open-Source Distributed Job Scheduler

<https://github.com/alexandra-mw/distributed-scheduler>

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- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 2: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
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### Real-Time Collaborative Document Engine with CRDT Conflict Resolution

<https://crdt-docs.example.com>

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- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.

### Project Without Any Description (edge case)

<https://no-description.example.com>

### Personal Site Generator

<https://github.com/alexandra-mw/site-gen>

- A small static-site generator written in Go with incremental builds and a plugin system.

## EDUCATION

### Massachusetts Institute of Technology

2009 - 2011

#### Master of Science in Computer Science and Artificial Intelligence

Cambridge, MA

- Thesis on fault-tolerant consensus in geo-distributed databases
- Graduate Teaching Assistant for Distributed Systems
- GPA 3.9 / 4.0

### University of California, Berkeley

2005 - 2009

#### Bachelor of Science in Electrical Engineering and Computer Science

Berkeley, CA

- Graduated summa cum laude
- President of the ACM student chapter

### Online Continuing Education (no highlights)

2020 - 2020

#### Professional Certificate in Machine Learning

## CERTIFICATIONS AND AWARDS

- AWS Certified Solutions Architect - Professional
- Google Cloud Professional Cloud Architect
- Certified Kubernetes Administrator (CKA)
- Certified Kubernetes Application Developer (CKAD)
- HashiCorp Certified: Terraform Associate
- Microsoft Certified: Azure Solutions Architect Expert
- Distinguished Engineering Award, 2023
- Hackathon Grand Prize Winner, Global Cloud Summit, 2022

## **SPEAKING AND PUBLICATIONS**

- Keynote: Designing Resilient Systems at Planet Scale - Global Cloud Summit 2023.
- Conference talk: The Pragmatic Path to Microservices - QCon San Francisco 2022.
- Co-author of a widely cited paper on consensus protocols in geo-distributed databases.
- Regular contributor to several popular open-source infrastructure projects.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing infrastructure cost by a measurable margin and improving p95 latency across critical paths.

## **LANGUAGES**

- English (Native)
- Spanish (Fluent)
- French (Professional)
- German (Conversational)
- Mandarin (Basic)

## **ADDITIONAL LINKS OR DATA**

**Portfolio:** <https://alexandra.dev>

**GitHub:** <https://github.com/alexandra-mw>

**Case Studies:** <https://averylongportfoliourlthatdoesnotbreaknicely.example.com/case-studies>

**X / Twitter:** @alexandra\_mw